PeerLearn: Software Requirements Specification (SRS)

PeerLearn SRS v1.1 defines 15 functional requirements, 8 non-functional requirements and 8 security requirements for a campus peer-tutoring and study-group matching platform. Team: Amogh Shetty (PES1UG24AM034), Aarush Patil (PES1UG24AM005), Abheek Sen (PES1UG24AM009), Allen Anish (PES1UG24AM029). Course: Software Engineering, PES University.

Revision history. v1.0: version submitted for Part 1. v1.1 (31 Oct 2026): team members added; section 2.3 states that each account holds one role and that Admin accounts are created by the system administrator; sections 2.4 and 2.5 add Docker, Docker Hub and Kubernetes.

1. Introduction

1.1 Purpose. This document states what PeerLearn shall do and how well, so the team, the faculty and the testers share one description of the product. It is written in the IEEE SRS structure (ISO/IEC/IEEE 29148, earlier IEEE 830).

1.2 Scope. PeerLearn is a web application where students offer help in subjects (Tutors), ask for help (Learners), get matched by a rule-based scoring method, book sessions and join study groups. It does not include video calling, online payments or integration with the university's timetable system.

1.3 Definitions, acronyms and abbreviations.

Term	Meaning
Learner	A student who asks for help in a subject
Tutor	A student who offers help in one or more subjects
Admin	A staff or student moderator who handles reports
Match score	A number from 0 to 100 that ranks how well a Tutor fits a help request
Session	A booked meeting between one Learner and one Tutor at an agreed slot
FR / NFR / SR	Functional, non-functional and security requirement
RBAC	Role-based access control
JWT	JSON Web Token, used to keep a user logged in
REST API	Web interface used by the frontend to talk to the

Term	Meaning
	backend

1.4 References. ISO/IEC/IEEE 29148:2018 (requirements engineering); IEEE 830-1998 (SRS practice); OWASP Top 10 (security guidance).

1.5 Overview. Section 2 describes the product in general terms. Section 3 lists the use cases, then the specific functional, non-functional and security requirements, each with a unique ID used later in the Test Plan and the Architecture document.

2. Overall description

2.1 Product perspective. PeerLearn is a new, standalone web system with three parts: a browser frontend, a backend REST API and a relational database. It sends email through an external email service.

2.2 Product functions. In summary: account management (FR-01, FR-02), Tutor profiles and availability (FR-03, FR-04), help requests and matching (FR-05 to FR-07, FR-15), session booking (FR-08), ratings (FR-09), study groups (FR-10), notifications (FR-11), moderation (FR-12), dashboard (FR-13) and privacy of contact details (FR-14).

2.3 User classes and characteristics.

User class	Description	Technical skill
Learner	Any registered student asking for help	Basic web use
Tutor	Any registered student offering help. Each account holds one role (FR-02), so a student who also wants help registers a second account	Basic web use
Admin	Reviews reports and suspends accounts. Admin accounts are created by the system administrator and cannot be self-registered	Basic web use

2.4 Operating environment. Latest versions of Chrome, Firefox, Edge and Safari on desktop and mobile. The backend runs on Node.js with a PostgreSQL database. All tiers are packaged as Docker containers and deployed on Kubernetes.

2.5 Design and implementation constraints. The project must be finished in two one-week sprints (19 to 23 Oct and 26 to 30 Oct 2026) by a team of four, using Git and GitHub for source control, GitHub Actions for automation, Docker and Docker Hub for packaging, and Kubernetes for deployment.

2.6 Assumptions and dependencies. All users have a PES University email address. An email delivery service is available. Users have internet access during use.

3. Specific requirements

3.1 External interface requirements.

- **User interface:** responsive web pages for Learner, Tutor and Admin (see NFR-05).
- **Software interfaces:** PostgreSQL database through SQL; an email service through SMTP.
- **Communication interfaces:** JSON over REST using HTTPS (see SR-01).
- **Hardware interfaces:** none beyond a device with a web browser.

3.2 Use cases. Figure 1 shows who uses PeerLearn and what each actor can do. The table below links each use case to the requirements that define it.

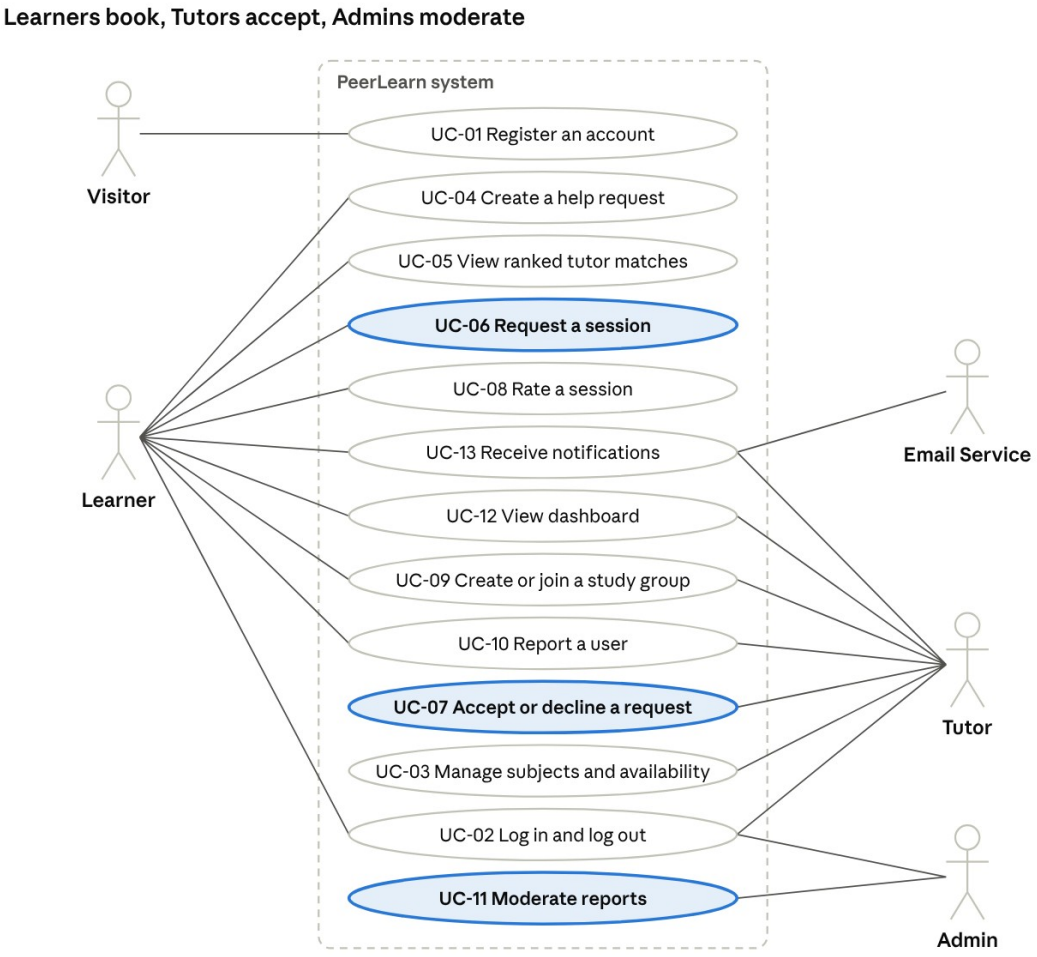


Figure 1: Use case diagram of PeerLearn. Learners request sessions, Tutors respond to them, and only Admins moderate reports.

ID	Use case	Actor	Requirements
UC-01	Register an account	Visitor	FR-01
UC-02	Log in and log out	Learner, Tutor, Admin	FR-02
UC-03	Manage subjects and availability	Tutor	FR-03, FR-04
UC-04	Create a help request	Learner	FR-05
UC-05	View ranked tutor matches	Learner	FR-06, FR-07, FR-15
UC-06	Request a session	Learner	FR-08, FR-14
UC-07	Accept or decline a session request	Tutor	FR-08, FR-14
UC-08	Rate a session	Learner	FR-09
UC-09	Create or join a study group	Learner, Tutor	FR-10
UC-10	Report a user	Learner, Tutor	FR-12
UC-11	Moderate reports	Admin	FR-12
UC-12	View dashboard	Learner, Tutor	FR-13
UC-13	Receive notifications	Learner, Tutor, Email Service	FR-11

3.3 Functional requirements. Priority: M = must have, S = should have. The last column names the test case in the Test Plan that verifies the requirement.

ID	Requirement	Priority	Verified by
FR-01	The system shall let a visitor register with a name, an email ending in @pes.edu and a password of at least 8 characters, and shall reject any other email domain with an error message.	M	TC-01, TC-02
FR-02	The system shall authenticate a user by email and password,	M	TC-03, TC-04

ID	Requirement	Priority	Verified by
	assign exactly one role (Learner, Tutor or Admin) and end the session on logout.		
FR-03	The system shall let a Tutor add, edit and delete up to 10 subjects, each with a proficiency level of Beginner, Intermediate or Advanced.	M	TC-05
FR-04	The system shall let a Tutor define weekly availability slots of 30 to 120 minutes and shall reject a slot that overlaps an existing slot.	M	TC-05
FR-05	The system shall let a Learner create a help request with a subject, a topic, a preferred date and time, and an urgency of Low, Normal or High.	M	TC-06
FR-06	The system shall calculate a match score from 0 to 100 for each Tutor teaching the requested subject, using subject match (50 points), availability overlap (30 points) and proficiency (20 points), and shall exclude Tutors who do not teach the subject.	M	TC-06
FR-07	The system shall show the Learner the 5 highest-scoring Tutors	M	TC-06

ID	Requirement	Priority	Verified by
	in descending order of score, each with their available slots.		
FR-08	The system shall let a Learner request a slot with a matched Tutor and let the Tutor accept or decline it, and shall expire a request that is not answered within 48 hours.	M	TC-07, TC-08
FR-09	The system shall let a Learner rate a completed session from 1 to 5 with an optional comment of at most 300 characters, once per session.	S	TC-09
FR-10	The system shall let a user create a study group for a subject with a maximum size of 3 to 10 members, and let Learners join or leave it until it is full.	S	TC-10
FR-11	The system shall send an in-app notification to the affected user when a session is requested, accepted or declined, and 1 hour before a session starts.	S	TC-07, TC-08
FR-12	The system shall let a user report another user with a reason, and shall let an Admin dismiss the report or warn or suspend the	S	TC-11

ID	Requirement	Priority	Verified by
	reported user.		
FR-13	The system shall show each user a dashboard with upcoming sessions and pending requests, and show a Tutor their average rating.	S	TC-17
FR-14	The system shall hide a user's email address and phone number from other users until the Tutor accepts a session request between them.	M	TC-07
FR-15	The system shall add up to 10 points to a Tutor's match score, calculated as average rating divided by 5 times 10, once the Tutor has at least 3 ratings, with the total capped at 100.	S	TC-06

3.4 Non-functional requirements. Each one is measurable, so a tester can say pass or fail.

ID	Category	Requirement	Verified by
NFR-01	Security	The system shall store passwords only as bcrypt hashes with a cost factor of at least 10.	TC-12
NFR-02	Performance	The system shall respond to 95% of API requests within 1 second with 100 concurrent users.	TC-18
NFR-03	Performance	The matching function shall return results within 2 seconds with	TC-13

ID	Category	Requirement	Verified by
		1,000 registered users and 200 Tutors in the database.	
NFR-04	Security	The system shall allow at most 100 API requests per minute per user and reject further requests with HTTP 429.	TC-14
NFR-05	Usability	The interface shall work without horizontal scrolling on screens 360 pixels wide and above, and every main function shall be reachable within 3 clicks from the dashboard.	TC-17
NFR-06	Auditability	The system shall record each login, booking and Admin action with user ID, timestamp and action, and keep the records for 90 days.	TC-16
NFR-07	Availability	The system shall be available at least 99% of the time between 08:00 and 22:00 IST, measured monthly.	Monitoring report
NFR-08	Maintainability	Automated unit tests shall cover at least 70% of backend code lines and shall run on every pull request.	CI coverage report

3.5 Security requirements. Objectives say what must be protected; requirements say exactly what the system shall do to protect it.

Security objectives

ID	Objective	Meaning for PeerLearn
SO-1	Confidentiality	Passwords, emails and phone numbers are visible only to those allowed to see them.
SO-2	Integrity and access control	Each user can do only what their role allows, and data cannot be changed through bad input.
SO-3	Accountability	Important actions can be traced to a user and cannot be erased.
SO-4	Availability and abuse resistance	The service stays usable and resists brute-force and flooding attempts.

Security requirements

ID	Requirement	Objective	Verified by
SR-01	The system shall transmit all data over HTTPS using TLS 1.2 or higher and reject plain HTTP.	SO-1	TC-19
SR-02	The system shall never include a password hash or token in any API response.	SO-1	TC-12
SR-03	The system shall return a user's email and phone number only to that user, to an Admin, or to the other party of an accepted session.	SO-1	TC-07
SR-04	The system shall require a valid token for every endpoint except register and login, expire tokens after 2 hours and answer an expired token with HTTP 401.	SO-2	TC-03, TC-19

ID	Requirement	Objective	Verified by
SR-05	The system shall answer a request to a role-restricted endpoint from a user of another role with HTTP 403.	SO-2	TC-04
SR-06	The system shall validate the type, length and format of all input on the server and shall use parameterised database queries.	SO-2	TC-15
SR-07	The system shall provide no function that edits or deletes audit log entries.	SO-3	TC-16
SR-08	The system shall lock an account for 15 minutes after 5 consecutive failed logins; NFR-04 supplies the request limit.	SO-4	TC-14